

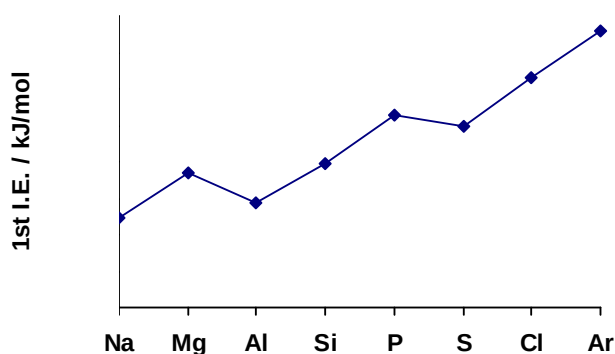
TPJC 2008 Prelim H1 Chem Paper 1: Multiple Choice Questions

1	A	6	C	11	D	16	A	21	A	26	B
2	B	7	D	12	C	17	C	22	C	27	B
3	C	8	A	13	B	18	C	23	A	28	C
4	D	9	A	14	C	19	C	24	B	29	A
5	D	10	C	15	C	20	B	25	D	30	D

TPJC 2008 Prelim H1 Chem Paper 2

Question 1

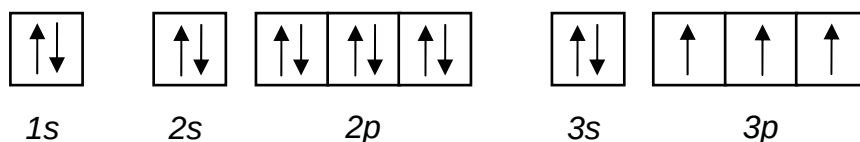
a) i)



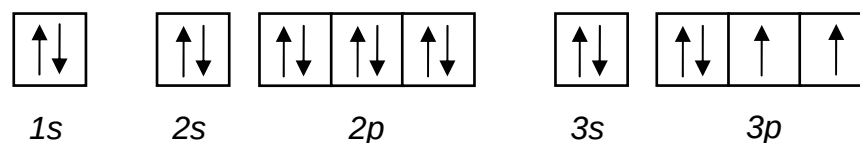
- ii) Across the period, nuclear charge increases due to increasing number of protons in the nucleus.
However, the screening effect remains similar across the period.
This results in the valence electron being more tightly held (or more energy is required for removal of valence electron).

b) i)

phosphorous



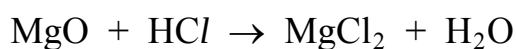
sulphur



- ii) In S, there is inter- electronic repulsion between the paired 3p electrons.
Hence less energy is required for the removal.

c) Magnesium oxide is basic.

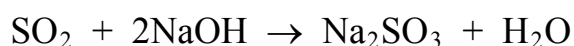
It reacts with acids to form salt and water.



(OR It dissolves in water to form an alkali. $\text{MgO} + \text{H}_2\text{O} \rightarrow \text{Mg(OH)}_2$)

Sulphur dioxide is acidic.

It reacts with bases to form salt and water.

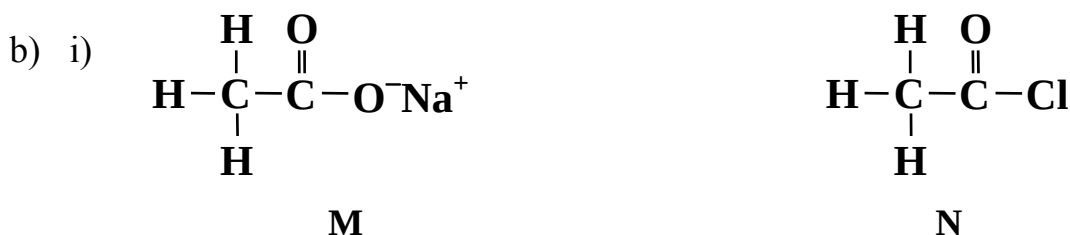


(OR It dissolves in water to form acids. $\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$)
 Also accept answers using SO_3 as the oxide of sulphur.

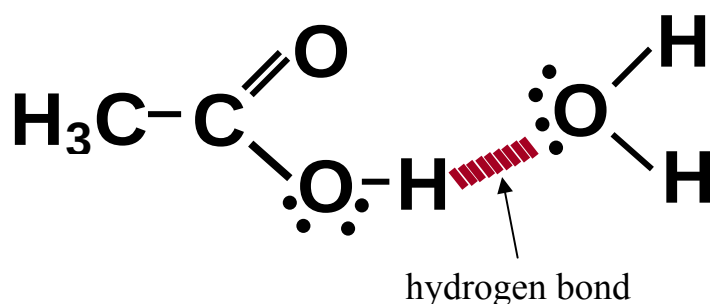
- d) i) $\Delta H = -908 - (-814) = \underline{-94 \text{ kJ mol}^{-1}}$
 ii) Energy evolved = 94 kJ = 94 000 J
 Mass of solution = $1.06 \times 1000 = 1060 \text{ g}$
 $94\,000 = 3.50 \times 1060 \times \Delta T \Rightarrow \Delta T = \underline{25.3 \text{ K}}$

Question 2

- a) i) A buffer solution can resist drastic changes in pH when a small amount of acid or base is added.
 ii) When a small amount of H^+ is added, it is removed by CH_3COO^- :
 $\text{CH}_3\text{COO}^- + \text{H}^+ \rightarrow \text{CH}_3\text{COOH}$
 When a small amount of OH^- is added, it is removed by CH_3COOH :
 $\text{CH}_3\text{COOH} + \text{OH}^- \rightarrow \text{CH}_3\text{COO}^- + \text{H}_2\text{O}$



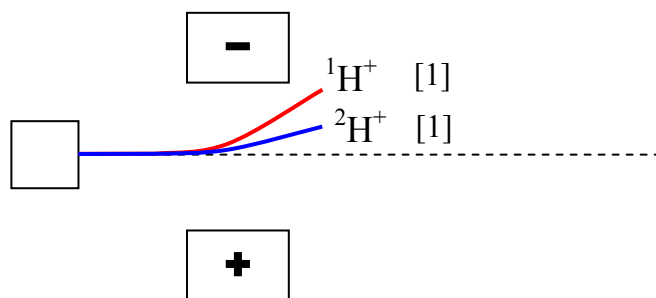
- ii) I : $\text{K}_2\text{Cr}_2\text{O}_7$ / acidified / warm
 II : Cl_2 (g) or Cl_2 (organic solvent) / uv or sunlight
 iii) Ethanoic acid is able to form hydrogen bonds with water.



- iv) pK_a of chloroethanoic acid is lower than that of ethanoic acid.
Cl atom is electron-withdrawing and disperses the charge on $\text{CH}_2\text{ClCOO}^-$, making $\text{CH}_2\text{ClCOO}^-$ more stable than CH_3COO^- .
 Thus CH_2ClCOOH dissociates more readily, making it a stronger acid.
 v) Fluoroethanoic acid would be a stronger acid as the F atom, being more electronegative, is more electron-withdrawing than Cl.

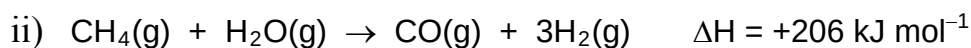
Question 3

a) i)



b) Deuterium has a heavier mass, requiring more energy to separate the molecules.

c) i) Methane has a high hydrogen to carbon ratio. (*in fact it is the highest ratio*)



Bonds broken

4 x C–H (4 x 410)

2 x O–H (2 x 460)

Bonds formed

C≡O

3 x H–H (3 x 436)

$$+206 = 4(410) + 2(460) - \text{B.E.}(\text{C}\equiv\text{O}) - 3(436)$$

$$\text{B.E.}(\text{C}\equiv\text{O}) = \underline{\underline{1046 \text{ kJ mol}^{-1}}}$$

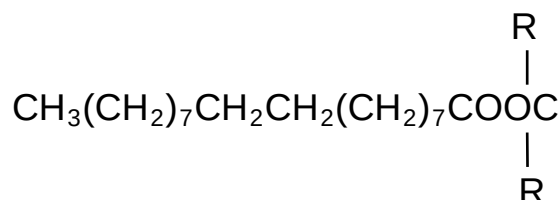
d) i) Moles of $\text{H}_2 = (3 \times 10^{12}) / 22.4 = \underline{\underline{1.34 \times 10^{11} \text{ mol}}}$

ii) Amount of energy required = $400 \times 1.34 \times 10^{11} = \underline{\underline{5.36 \times 10^{13} \text{ kJ}}}$



e) i) presence of Ni at 150°C (or Pt catalyst)

ii)



(accept answer with 2 side-chains hydrogenated)

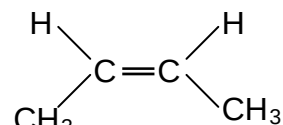
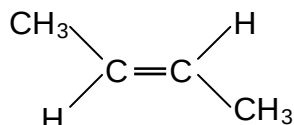
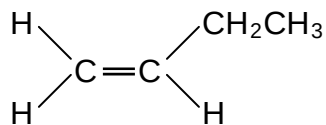
iii) There is restricted rotation about the C=C bond.

Section B

Question 4

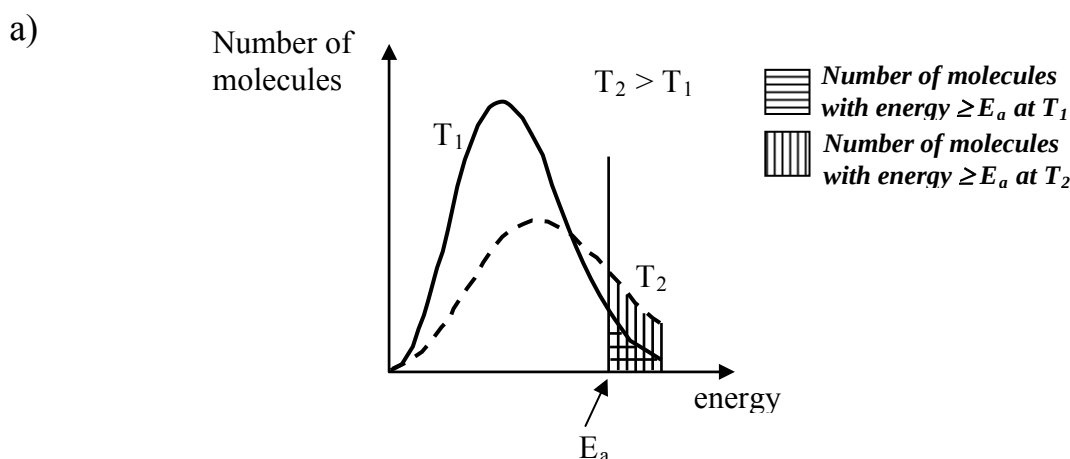
a) i) alcoholic KOH, reflux

ii)



- b) i) **A** $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$
B $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
C $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 ii) Rate of reaction for bromopropane is faster.
 C–Br bond is weaker than C–Cl bond, therefore more easily broken.
- c) i) HCl (g) (or PCl_3 or PCl_5 or SOCl_2)
 ii) acidified $\text{K}_2\text{Cr}_2\text{O}_7$ (or acidified KMnO_4) / heat
 iii) excess concentrated H_2SO_4 / reflux
- d) The intermolecular forces of attraction in butan-1-ol is hydrogen bonding whereas that in 1-chlorobutane is permanent dipole-dipole attraction.
 Since hydrogen bonding is stronger than permanent dipole-dipole attraction, more energy is required to overcome the intermolecular forces (or separate the molecules.)
- e) i) $\text{Ca(OH)}_2 + 2\text{CH}_3\text{COOH} \rightarrow \text{Ca(CH}_3\text{COO)}_2 + \text{H}_2\text{O}$
 ii) Empirical Formula : $\text{C}_3\text{H}_6\text{O}$
 $M_r \text{ of } (\text{C}_3\text{H}_6\text{O})_n = 58 \Rightarrow n=1$
D is propanone (or CH_3COCH_3)
 iii) Test 1:
 Add 2,4-DNPH.
 Observation: An orange ppt obtained would show that **D** is a carbonyl.
 Test 2 :
 Warm **D** with Tollen's reagent. (Fehling's test not accepted)
 Observation: No silver mirror obtained would confirm that **D** is a ketone.

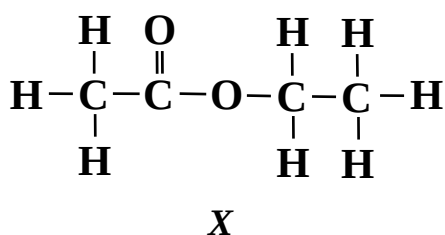
Question 5



When temperature is raised average kinetic energy of molecules increases.
 The number of molecules which possess K.E. $\geq$ activation energy (E_a) is increased.
 This results in greater number of effective collisions.

Thus rate is increased when temperature is increased.

b) i)

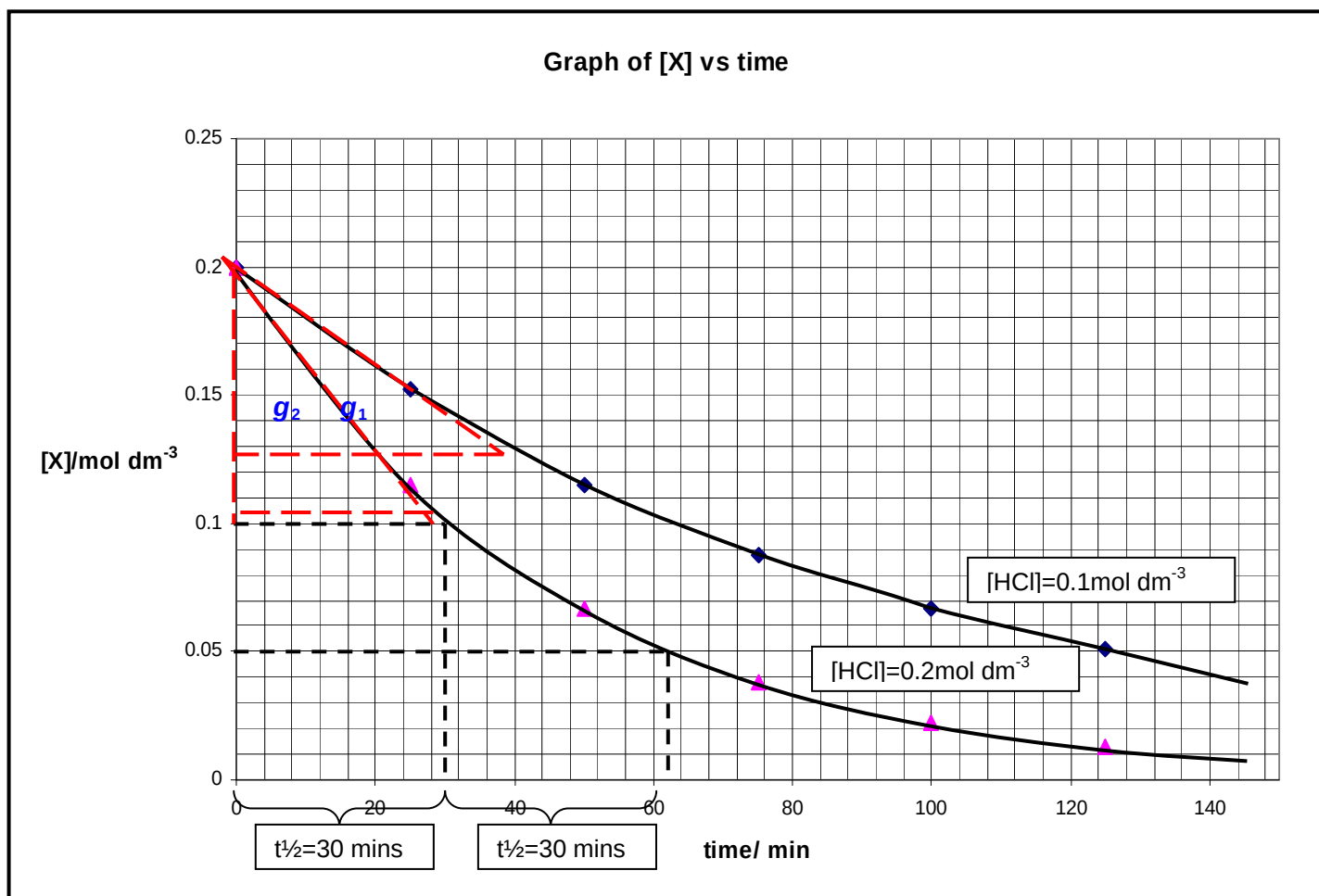


ii) Shape of graphs

[1]

Axes Labels

[1]



iii) From the graph, half life is constant.

[working shown on graph for 1 of the curves]

→ the **order of reaction wrt [X] is 1.**

By using initial rates method,

$$\text{For } [\text{HCl}] = 0.1 \text{ mol dm}^{-3}, \text{ gradient, } g_1 = \frac{0.075}{40} = 1.88 \times 10^{-3}$$

For $[\text{HCl}] = 0.2 \text{ mol dm}^{-3}$, gradient, $g_2 = \frac{0.1}{28} = 3.57 \times 10^{-3}$

When $[\text{HCl}]$ is doubled, the rate is also doubled.

→ the **order of reaction wrt $[\text{HCl}]$ is 1.**

iv) $\text{rate} = k[\text{X}][\text{HCl}]$ or $\text{rate} = k[\text{CH}_3\text{COOCH}_2\text{CH}_3][\text{HCl}]$

Using data from $[\text{HCl}] = 0.2 \text{ mol dm}^{-3}$ curve,

$$k = \frac{3.57 \times 10^{-3}}{(0.2)(0.200)} = \mathbf{0.0892 \text{ dm}^3 \text{ mol}^{-1} \text{ min}^{-1}}$$

$[\frac{1}{2}] \qquad \qquad [\frac{1}{2}]$

- c) i) NaCl has ionic structure consisting of ions held together by strong electrostatic forces of attraction.
Both AlCl_3 and SiCl_4 have a simple molecular structure consisting of molecules held together by weak induced dipole-dipole attraction.
Since the electrostatic attraction is stronger than the induced dipole-dipole attraction, more energy is required to overcome the strong attraction between ions in NaCl .
 AlCl_3 can exist as Al_2Cl_6 dimers.
The larger number of electrons in the Al_2Cl_6 molecules result in stronger induced dipole-dipole attraction compared to SiCl_4 , therefore requiring more energy to overcome the attraction.
- ii) $[\text{Al}(\text{H}_2\text{O})_6]^{3+}(\text{aq}) \longrightarrow [\text{Al}(\text{H}_2\text{O})_5(\text{OH})]^{2+}(\text{aq}) + \text{H}^+(\text{aq})$
 $\text{pH} = 3$

Question 6

- a) i) $\text{CrO}_4^{3-} + 8\text{H}^+ + 2\text{e}^- \rightarrow \text{Cr}^{3+} + 4\text{H}_2\text{O}$
 $2\text{CrO}_4^{3-} + 2\text{H}^+ \rightarrow \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O} + 2\text{e}^-$
- ii) $3\text{CrO}_4^{3-} + 10\text{H}^+ \rightarrow \text{Cr}^{3+} + \text{Cr}_2\text{O}_7^{2-} + 5\text{H}_2\text{O}$
Disproportionation refers to a reaction in which the same element undergoes both oxidation and reduction.
- iii) moles of $\text{S}_2\text{O}_3^{2-} = 0.04 \times 0.25 = 0.01 \text{ mol}$
moles of $\text{Cr}_2\text{O}_7^{2-} = 0.01/6 = 0.00167 \text{ mol}$
moles of $\text{CrO}_4^{3-} = 3 \times 0.00167 = 0.00501 \text{ mol}$
mass of $\text{Na}_3\text{CrO}_4 = 0.00501 \times 185 = \mathbf{0.927\text{g}}$
- b) i) By Le Chatelier's principle, decrease in temperature favours the exothermic reaction as the system tries to produce more heat.
Since the forward reaction is exothermic, the equilibrium will shift right, thus increasing the yield of ammonia.
- ii) pressure of $\sim 250 \text{ atm}$ / temperature of $\sim 450^\circ\text{C}$

Other factors:
 rate of reaction
 cost of operating at high pressure

iii)

	N ₂ (g)	H ₂ (g)	NH ₃ (g)
No of moles at eqm / mol	2.86 x 10 ⁻²	4.55 x 10 ⁻²	2.15 x 10 ⁻²
Vol of container / dm ³	0.5 dm ³		
Conc at eqm / mol dm ³	5.72 x 10 ⁻²	9.10 x 10 ⁻²	4.30 x 10 ⁻²

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

$$= \frac{(4.30 \times 10^{-2})^2}{(5.72 \times 10^{-2})(9.10 \times 10^{-2})^3}$$

$$= \underline{\underline{42.9 \text{ mol}^{-2} \text{ dm}^6}}$$

c) i) pH is a measure of the acidity or alkalinity of a solution and is defined as

$$\text{pH} = -\log_{10} [H^+]$$

ii) $[H^+] = 10^{-2.35} = \underline{\underline{4.47 \times 10^{-3} \text{ mol dm}^{-3}}}$

iii) Amount of NaOH = 0.0285 x 0.10 = 2.85 x 10⁻³ mol

Mole ratio of NaOH : acid

1 : 1

Molar concentration of lime juice = 2.85 x 10⁻³ / 0.025 = **0.114 mol dm⁻³**

iv) The acid is a weak acid and is not fully ionised.

v) Phenolphthalein is the suitable indicator as this is a strong base-weak acid titration and the end-point pH falls within phenolphthalein's working range (or in the alkaline region).